Skills Summary

Diagnosing an Overgeneralized Number-Pattern Rule

Use this reference while completing your worksheet or when studying.

Skill 1 — Test the rule on terms nobody checked

Rule: a rule that fits the terms you used to *make* it is not evidence. Test it on a term you did not use.

- Write the step between every pair of terms you can see, not just the first pair.
- If all the steps are the same number, the rule is "add that step".
- If each term is the one before times the same number, the rule is "multiply".
- "Add" and "multiply" rules often agree on the first two terms — that agreement proves nothing.

Example: 2, 4, 6, 8, . . . Someone says "double the term before". Test it, then find the 9th term.
Step 1. Doubling and adding both explain $2 \to 4$, so test the next step instead: doubling 4 predicts 8, but the third term is 6 — the doubling rule is out.
Step 2. Every step is $+2$, so the nth term is $2 \times n$, and the 9th term is 2×9. The 9th term is **18**.

Try it: 7, 14, 21, 28, . . . Test "double the term before", then find the 10th term.

check: **70**

Skill 2 — Beat a rule with one counterexample

Rule: to disprove "*every* A is B", you need one A that is not B. One counterexample beats any number of supporting examples.

Testing "is it a multiple?": divide and look at the remainder. A remainder of 0 means yes; any other remainder means no — and that number is your counterexample.

Example: "Every multiple of 3 is a multiple of 9." Test the multiple of 3 that the claim never mentions: 15.
Step 1. To disprove a "multiple of" claim, divide the counterexample candidate by 9 and look at the remainder rather than hunting for more examples that work.
Step 2. Nine goes into fifteen once: $15 - 9 \times 1$ is what is left over. The remainder is **6**, so 15 is not a multiple of 9 and the claim is false.

Try it: "Every multiple of 5 is a multiple of 10." Divide 25 by 10 and write the remainder.

check: **5**

Skill 3 — Repair the rule, then use it

How to repair an overgeneralized rule:

- Split the claim into parts and test each part separately — often one half survives and one half fails.
- Keep the part every term supports; cross out the part one term refutes.
- Write the repaired rule as a shortcut you can jump with: the multiples of k have nth term $k \times n$.

Example: 6, 12, 18, 24, ... Claim: "the terms are the multiples of 6, so every term is even and ends in 6." Repair it, then find the 11th term.
Step 1. Test the claim one half at a time: "multiples of 6" holds for every term, but "ends in 6" already fails at the second term, so only the first half is kept.
Step 2. The repaired rule gives the nth term as $6 \times n$, so jump straight to 6×11 instead of listing terms. The 11th term is **66**.

Try it: 9, 18, 27, ... Repair the claim "every term is a multiple of 9 and ends in an odd digit", then find the 12th term.

check: **108**

(!) Watch out: "It worked three times" is the trap. Three cases can all pass while the rule is still false — and the cases people pick are usually the ones they built the rule from.